Supplementary Material

Label-Space Mismatch Distorts Cross-Cohort Evaluation in Breast Cancer Molecular Subtyping

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S1. Transformer Architecture Investigation

This supplementary section provides detailed results for the Transformer architecture investigation described in Section 7 of the main manuscript.

S1.1 Capacity Sweep

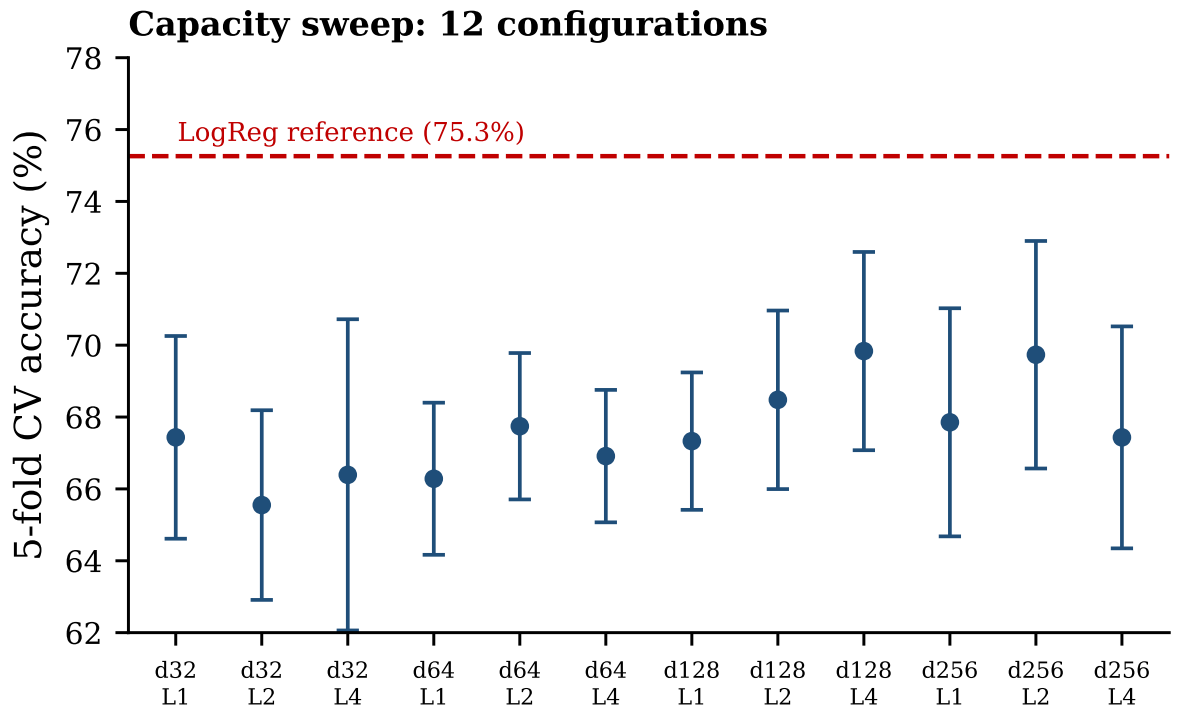


Figure 1: Capacity sweep over 12 configurations ($d \times L$). Every configuration lies below the logistic-regression reference (dashed), and eleven of twelve fall within a 4-point band, indicating that capacity is not the binding constraint.

S1.2 Learning Curves

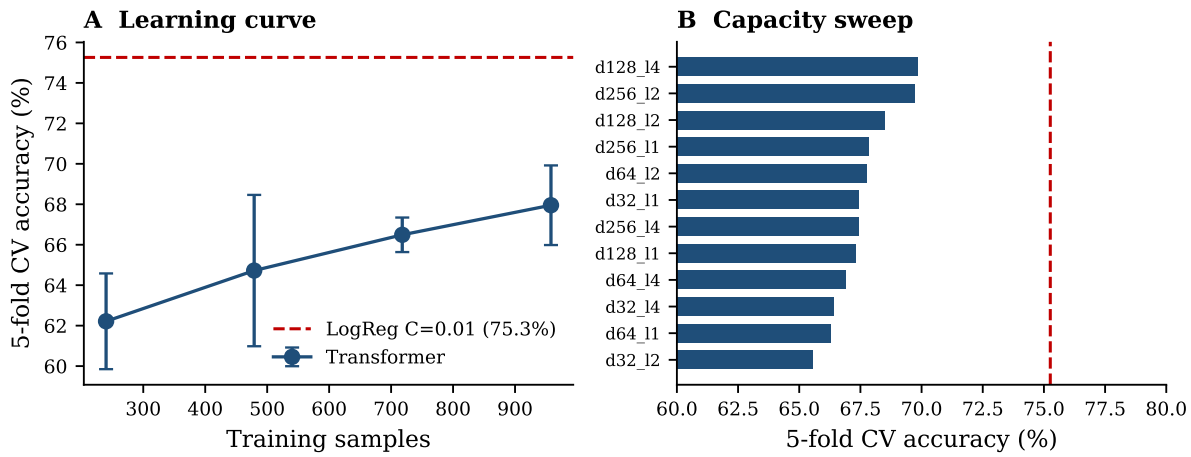


Figure 2: Learning curve for the best small configuration, with the logistic-regression reference. The curve is decelerating; extrapolation places parity at roughly six times the present training set.

S1.3 Regularization Sweep

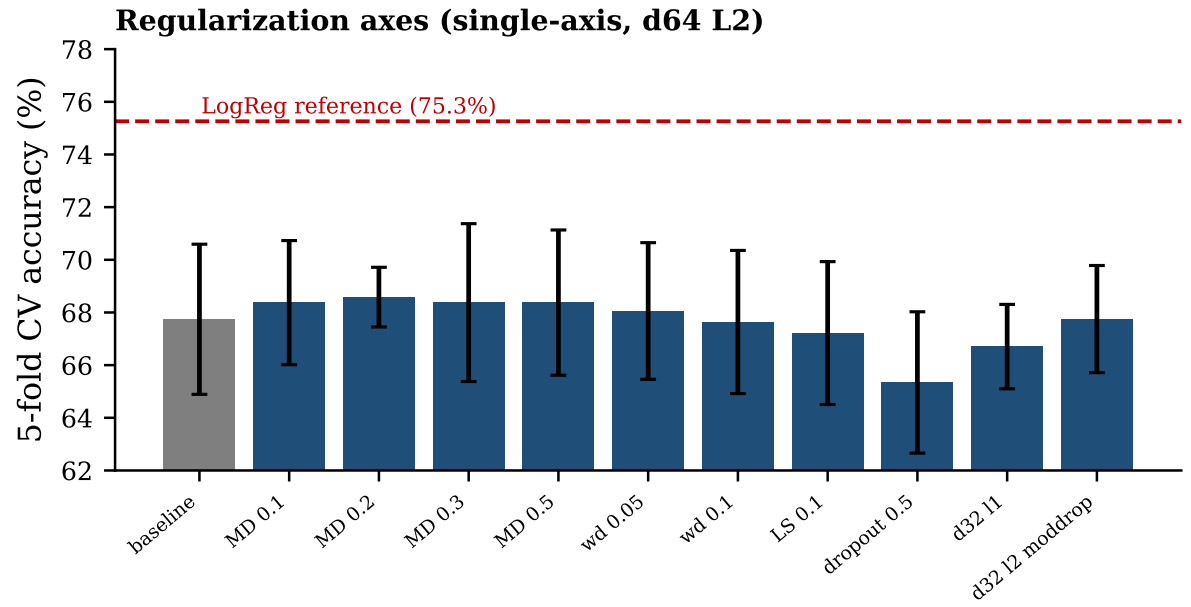


Figure 3: Single-axis regularization sweep. The strongest intervention gains 0.84 points, and no configuration approaches the logistic-regression reference.

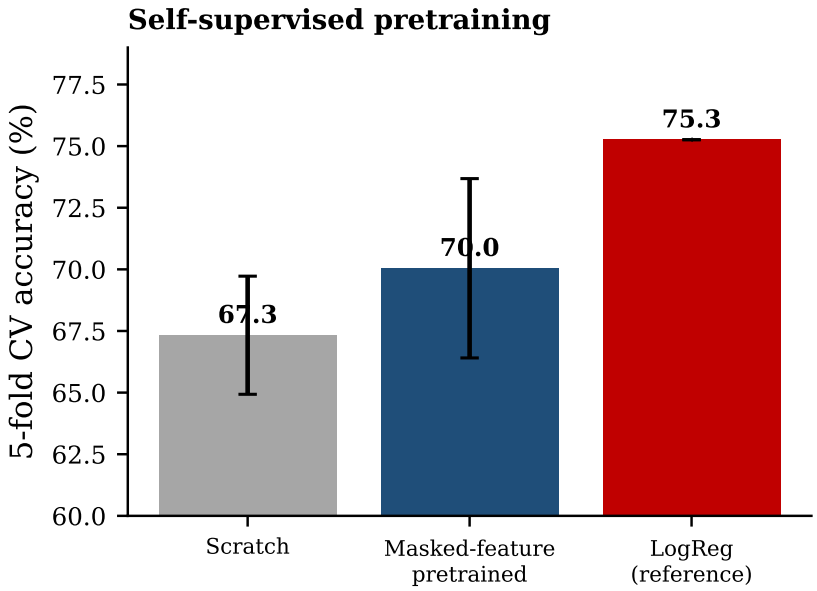


Figure 4: Masked-feature pretraining raises cross-validated accuracy by 2.7 points, the largest internal gain of any intervention tested, without reaching the linear reference.

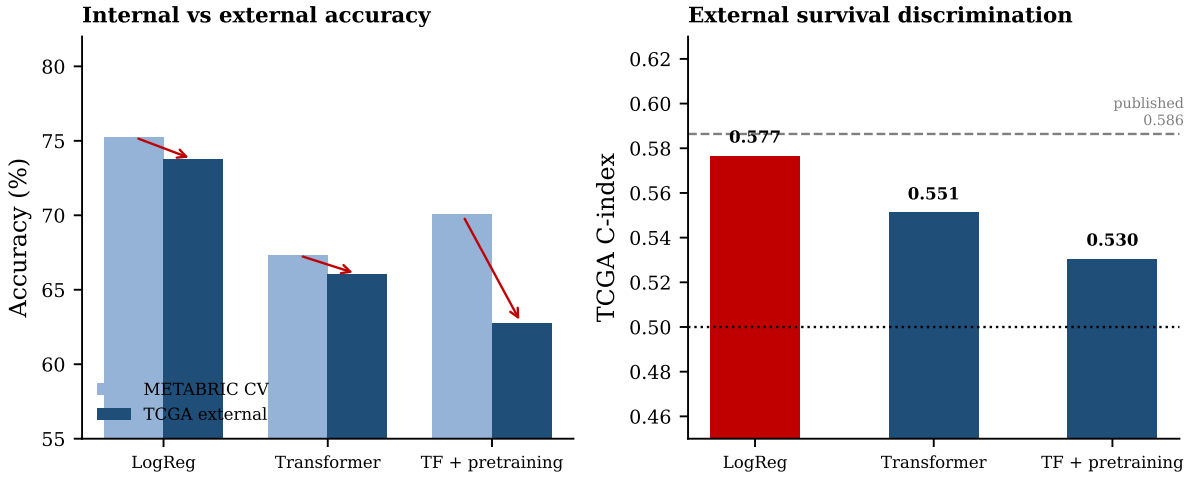


Figure 5: Internal versus external performance. Left: accuracy on METABRIC cross-validation and on TCGA. Right: external survival C-index. The pretrained Transformer shows the largest internal-to-external drop, and the lowest external C-index.

S1.4 Pretraining Results

S1.5 External Validation Reversal

S1.6 Supplementary Tables

Table 1: Complete feature engineering details for all five modalities.

Modality	**Variables**	**Encoding**	**Dimensions**
Clinical	22 variables	One-hot + Z-score	69
mRNA	412 genes	Z-score	412
CNA	256 genes	Z-score	256
Methylation	232 probes	Z-score	232
Mutation	173 genes	Binary	173
Total	**1,095**		**1,142**

Table 2: ANOVA feature selection results for mRNA genes.

Gene	**F-statistic**	**Rank**
ACOT4	130.8	1
A2ML1	102.0	2
ABAT	98.5	3
⋮	⋮	⋮

S2. Five-Class Training Baselines

The main text trains a six-class METABRIC model and reconciles its output space at evaluation time. An alternative is to train directly on five classes. We built both five-class variants from the same 412-gene rank-normalised mRNA features and the same C=0.01 logistic regression: *drop* removes the 91 claudin-low training samples (n=867), and *merge* relabels them as Basal-like (n=958). Both are evaluated on the same external cohorts as the six-class estimator.

Table 3: Five-class training baselines against the six-class estimator with label-space reconciliation. All rows use one rank-normalised mRNA feature set and logistic regression (C=0.01); rows differ only in training label space and post-hoc handling. ECE is computed on top-1 confidence with 15 bins; the Brier score is the multiclass sum-of-squares.

Cohort	Estimator	Acc (%)	Macro F1	ECE	Brier
TCGA-BRCA (n=987)					
6-class base		73.76	0.585	0.0352	0.3756
6-class mask_only		79.74	0.757	0.0557	0.2906
6-class em_pam50		**80.65**	0.737	0.0460	**0.2745**
5-class drop (n=867)		80.04	**0.774**	0.0560	0.2892
5-class merge		76.90	0.695	0.0344	0.3284
GSE21653 (n=266)					
6-class base		63.16	0.504	0.0836	0.4920
6-class mask_only		68.42	0.626	0.0364	0.4271
6-class em_pam50		**74.06**	0.683	0.0595	**0.3776**
5-class drop (n=867)		69.55	**0.643**	0.0467	0.4025
5-class merge		72.18	0.654	0.0481	0.3747

Two observations follow directly. First, on TCGA-BRCA the five-class *drop* model reaches 80.04% against 80.65% for the six-class estimator with EM correction, and the paired McNemar test does not separate them (p=0.504): for this cohort, training on five classes is an equally good route to the same accuracy, and it attains the highest macro F1 of any row (0.774). Second, on GSE21653 the six-class estimator with EM correction is significantly better than the five-class drop model (74.06% against 69.55%, p=0.0033), because EM additionally re-estimates the target class priors, which the basal-enriched composition of that cohort warrants. The *merge* variant is worse than both on TCGA (p=$8.0e-06$ against the six-class estimator) and statistically indistinguishable from it on GSE21653 (p=0.23).

The honest reading is that label-space *reconciliation* accounts for most of the recovered accuracy on TCGA, and prior correction adds materially only on the harder cohort. This does not weaken the main claim – the reported six-class score of 73.76% is still an artefact of the evaluation protocol – but it does mean that a five-class training run is a legitimate alternative when the target label space is known *a priori*.

S3. Calibration of the Corrected Estimates

One objection to prior correction is that it may only redistribute probability mass towards the dominant class without producing better probability estimates. The ECE and Brier columns of Table 3 test that directly, on the same predictions. The multiclass Brier score improves

monotonically along base → mask_only → em_pam50 on both cohorts (TCGA: 0.3756 → 0.2906 → 0.2745; GSE21653: 0.4920 → 0.4271 → 0.3776). The corrected probabilities are therefore closer to the observed outcomes in a proper scoring sense; the correction is not a pure relabelling of the argmax. ECE moves in the opposite direction at the mask step on TCGA (0.0352 → 0.0557) because restricting the label space sharpens the confidence distribution while the residual error is concentrated in the remaining classes; this is why we report both metrics rather than either alone.

S4. Paired Significance Tests

Table 4 reports McNemar tests with continuity correction between successive estimator stages, on the identical target samples.

Table 4: McNemar tests (continuity-corrected) between estimator stages. b is the number of samples the first estimator gets right and the second gets wrong; c is the converse.

Cohort	**Comparison**	b	c	χ^2	p
TCGA-BRCA	base_vs_mask	0	59	57.02	$4.3e-14$
TCGA-BRCA	mask_vs_em	20	29	1.31	0.2531
TCGA-BRCA	base_vs_em	13	81	47.76	$4.8e-12$
TCGA-BRCA	sixclass_em_vs_fiveclass_drop	31	25	0.45	0.5040
TCGA-BRCA	sixclass_em_vs_fiveclass_merge	51	14	19.94	$8.0e-06$
GSE21653	base_vs_mask	0	14	12.07	$5.1e-04$
GSE21653	mask_vs_em	1	16	11.53	$6.9e-04$
GSE21653	base_vs_em	1	30	25.29	$4.9e-07$
GSE21653	sixclass_em_vs_fiveclass_drop	13	1	8.64	0.0033
GSE21653	sixclass_em_vs_fiveclass_merge	8	3	1.45	0.2278

Two conclusions hold on both cohorts. The label-space step is unambiguous: it fixes 59 samples on TCGA and 14 on GSE21653 while breaking none, and both tests are decisive (p=4.3×10^{-14} and p=5.1×10^{-4}). The subsequent EM prior correction is significant on GSE21653 (p=6.9×10^{-4}) but not on TCGA (p=0.25), consistent with the accuracy table: prior correction pays off where the target composition departs substantially from the source.

S5. Generalising the Absence Pattern Beyond Claudin-Low

The analysis in the main text is built on one absent class, claudin-low, in two PAM50-annotated cohorts. A fair objection is that claudin-low happens to be the rarest class in the source prior,

so the reported effect may be the smallest instance of a larger phenomenon rather than a representative one. This section removes that ambiguity by sweeping absence over two axes: *who* is absent, and *how* they are absent. The protocol is otherwise the one described in the Methods: the same C=0.01 logistic regression on the 412-gene rank-normalised mRNA panel, a synthetic target of n=1000 drawn from the pooled held-out METABRIC validation and test splits, and 10 seeds per cell. METABRIC source priors are LumA

S5.1 Who is absent: severity tracks prior mass, not identity

Table 5: Single-class absence, complete (*drop*, target prevalence exactly zero) and partial (*rare*, target prevalence scaled to 10% of source). Rows are ordered by descending source prevalence. Bold marks the two commonest classes. Detection F1 is the label-free procedure's agreement with the true absent set.

		drop			rare	
Absent class	**Source prior**	**Naive**	**Corrected**	**Gain**	**Corrected**	**Det. F1**
LumA	0.357	74.67	86.00	+11.33	75.97	0.000
LumB	0.239	78.10	88.51	+10.41	85.93	0.067
Her2	0.125	78.82	81.51	+2.69	82.53	0.900
Basal	0.104	79.25	80.30	+1.05	79.89	0.900
CL	0.095	79.17	80.91	+1.74	79.85	0.967
Normal	0.079	79.34	81.51	+2.17	81.08	0.800

The result is unambiguous. Removing **Luminal A**, the commonest class, costs 11.33 points of naive accuracy and all of it is recoverable (74.67% $\rightarrow$ 86.00%). Removing claudin-low costs 1.74 points. The severity of a label-space mismatch is therefore governed by the *prior mass* of the absent class, not by its identity: claudin-low is the mildest instance of the phenomenon, not a representative one, and a cohort that omits a common subtype would show an effect an order of magnitude larger.

The partial-absence column (*rare*) exposes a second, less comfortable result. When the class is under-ascertained rather than absent outright, the label-free detector essentially fails for the common classes: detection F1 is **0.000** for Luminal A and **0.067** for Luminal B, against 0.967 for claudin-low. A class that is merely heavily under-counted leaves a probability marginal indistinguishable from a genuinely rare class, which is precisely the identifiability failure the main text reports, now shown to be worst exactly where the stakes are highest.

S5.2 How it is absent: merging is worse than dropping

Real annotation disagreement rarely takes the form of a class vanishing. More often a cohort folds one class into a neighbour's label – claudin-low recorded as Basal-like, or Luminal B recorded as Luminal A. We simulate this by moving the absent class's prior mass onto the neighbour instead of discarding it.

Table 6: Merge-mode absence: the source class is relabelled as its neighbour. Naive accuracy is barely affected, because the merge is partly *correct* under the target's coarser standard; the recoverable loss is correspondingly larger than for a clean drop.

Pattern	Absent class	Source prior	Naive	Corrected	Gain
LumA→LumB	LumA	0.357	76.73	89.82	+13.09
LumB→LumA	LumB	0.239	79.43	91.48	+12.05
Her2→LumB	Her2	0.125	79.04	81.18	+2.14
CL→Basal	CL	0.095	78.39	80.38	+1.99
Normal→LumA	Normal	0.079	79.36	82.04	+2.68

Luminal B merged into Luminal A – the exact pattern a reader is most likely to meet in practice – leaves naive accuracy at 79.43% while hiding a recoverable loss of 12.0 points. This is the most consequential cell in the whole simulation, and it involves neither a rare class nor a class vanishing: two subtypes with the same receptor status and overlapping expression, recorded under one label. Detection F1 here is 0.767, so the label-free procedure does not rescue it.

S5.3 Multiple absences

Dropping every pair of classes (15 combinations) gives a recoverable loss between +1.41 and +17.31 points, and the ordering again follows the combined prior mass: the worst cell is Luminal A with Luminal B (+17.31 points, naive accuracy 66.01%), the mildest is claudin-low with Basal-like (+1.41). Detection F1 is high across most pairs (1.000 in 11 of 15 cells) because two simultaneously suppressed classes produce a stronger simplex signal than one; the failures are concentrated in the pairs whose members are individually hardest to separate (claudin-low with Basal-like, 0.860).

Taken together, Supplementary Tables 5 and 6 support the main text's recommendation and sharpen it. The rule "establish the target's allowed labels from cohort documentation" is not a precaution against an exotic failure mode. It is the only reliable remedy for a mismatch whose magnitude ranges over two orders of magnitude depending on which class the annotation standard omits, and whose most likely real-world form – a class merged into its neighbour – is

the one that hides the most.